

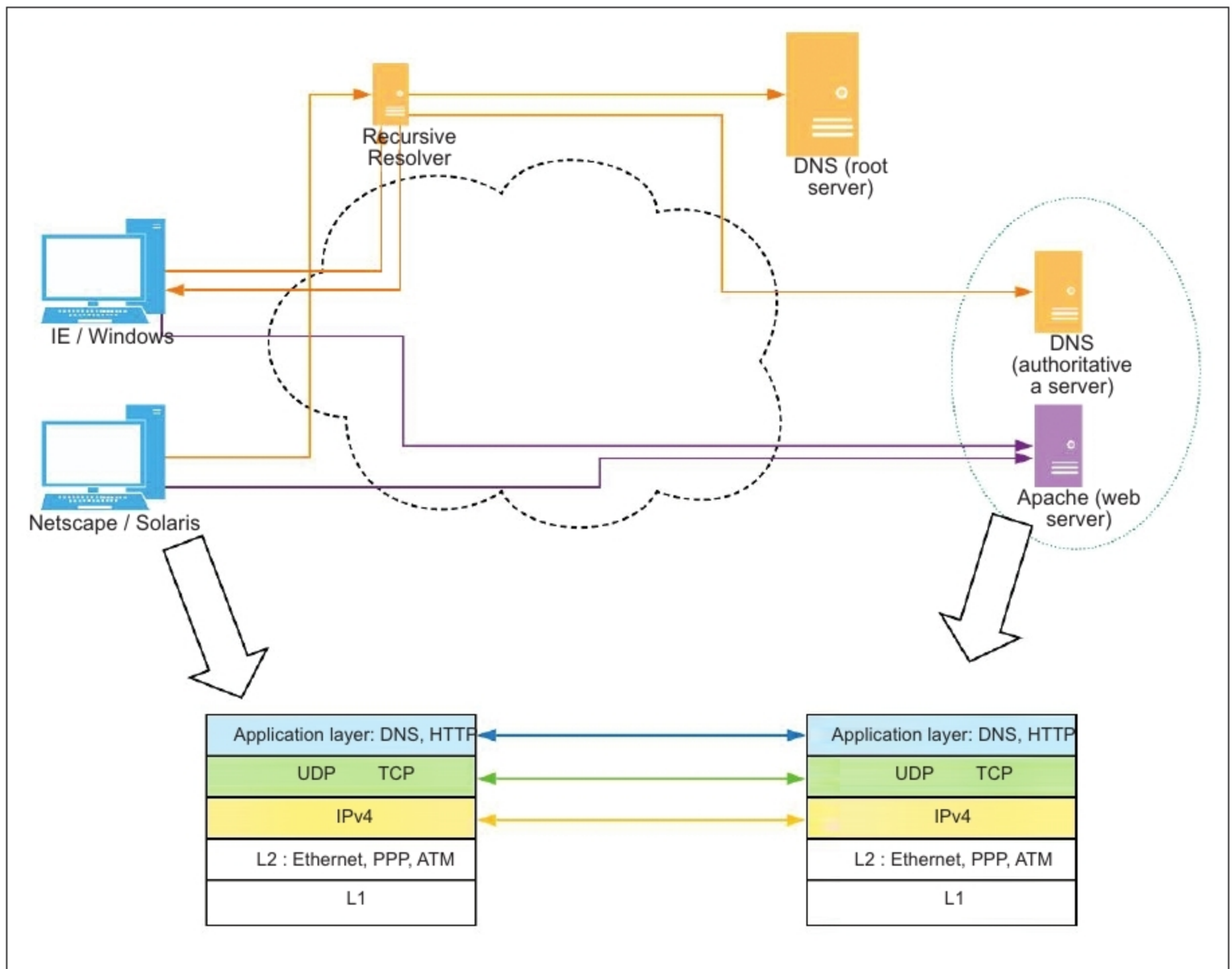


Figure 2: Web request with protocol stack

1. A user enters a URL in the browser or clicks a link; say, the destination URL is *http://www.example.com/test*. In this case, the sub-string *www.example.com* is referred to as the hostname (or, more formally, as FQDN or Fully Qualified Domain Name). In either case, the browser needs to connect to the Web server for the given URL, obtain the content and render it.
 2. To connect to the Web server, the browser requires a network address for the given hostname (in this case, *www.example.com*). This is achieved by performing a DNS lookup, which itself involves other sub-steps, which finally lead to locating a DNS server that returns the required network address. Such a server is called the authoritative server or nameserver for the FQDN.
 3. Once the network address is obtained, the browser connects with the Web server and initiates an HTTP request for the URI */test*. If the server returns data associated with the URI */test*, the browser infers the content type and renders it accordingly.
- Note the distinction between the complete URL, hostname and the URI. The end user is always interested in the complete URL but the communication system handles this in two parts: the hostname and the URI. It is worth

reiterating that the hostname is looked up over DNS to obtain the IP of the Web server. Once a connection is made with this IP address, only the URI (i.e., the component after the hostname) is sent over this connection.

Both steps 2 and 3 above involve the 'protocol stack'. This is shown in Figure 2, though for brevity, only the browser and Web server stacks are shown.

Figure 2 represents the core software protocols and layers of network communication. Also, the entities for browsers and Web servers are intentionally chosen to represent the early software. This leads us to the history of the protocol stack.